how should i code my route fuctions

For PavePlanner, I recommend coding each route as a function that displays a page or performs an action. Since you're using Drafter, each route will have a `@route(...)` decorator and receive `state` as a parameter.

Below is a starter structure for your 8 routes. It keeps estimates in memory only while the app is running.

The official Drafter pattern is to return `Page(state, [...])` from each route. Buttons connect pages by naming the route they should open, and form inputs send their values to matching route parameters.

Drafter Documentation

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1. Start with your dataclasses

Put these at the top of your Python file:

```
from drafter import *
from dataclasses import dataclass, field



@dataclass
class PavingProject:
    name: str
    length: float
    width: float
    price_per_sqft: float
    estimated_cost: float



@dataclass
class State:
    estimates: list[PavingProject] = field(default_factory=list)
    project_name: str = ""
    length: str = ""
    width: str = ""
```

```
    price_per_sqft: str = ""
    current_estimate: PavingProject | None = None
    selected_index: int = 0
```

2. Code your eight routes

These examples use the Drafter components `Header`, `Text`, `TextBox`, and `Button`.

They are a beginner-friendly starting point; you can improve the page design later.

```
# 1. HOME PAGE
@route
def home_page(state: State) -> Page:
    return Page(state, [
        Header("PavePlanner"),
        Text("Estimate the cost of your paving project."),
        Button("Create an Estimate", "new_estimate_page"),
        Button("View Estimates", "estimates_page")
    ])


# 2. NEW ESTIMATE PAGE
@route
def new_estimate_page(state: State) -> Page:
    return Page(state, [
        Header("Create a New Estimate"),
        "Project name:",
        TextBox("project_name"),
        "Length in feet:",
        TextBox("length", kind="number"),
        "Width in feet:",
        TextBox("width", kind="number"),
        "Price per square foot:",
        TextBox("price_per_sqft", kind="number"),
        Button("Calculate Estimate", "calculate_estimate"),
        Button("Home", "home_page")
    ])


# 3. CALCULATE ESTIMATE
@route
```

```
def calculate_estimate(state: State, project_name: str,
                       length: float, width: float,
                       price_per_sqft: float) -> Page:

   state.project_name = project_name
```

3. Two important changes

Before running the code, make these adjustments:

1. Make the home page the starting page. Drafter expects the main route to be named `index`. Rename `home_page` to `index`, and change every `Button("Home", "home_page")` to `Button("Home", "index")`.
2. Add a reset button. In `estimates_page`, add this line to the `content.extend([...])` list:

```
Button("Reset All Estimates", "reset_estimates"),
```

4. How the routes connect

Home (`index`)

New Estimate

Calculate Estimate

Estimates

View Details

Compare

Delete

Reset All

One thing to remember: Your estimates are stored in `state.estimates`, so they stay available as you navigate around the app during the session. They are not permanently saved.

I would build and test the routes one at a time rather than trying to debug all eight at once. The official

[Drafter documentation]

has examples of routes and form inputs.

can you make the compare estimates page create a graph comparing the estimates

Yes! You can use Matplotlib to create a bar graph comparing the estimated costs of all the projects created during the current session.

Drafter supports Matplotlib through its `MatPlotLibPlot()` component.

Drafter Documentation

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1. Add this import at the top of your code

```
import matplotlib.pyplot as plt
```

2. Replace your existing `compare_estimates_page` route with this

```
@route
def compare_estimates_page(state: State) -> Page:
    content = [Header("Compare Estimates")]

    if len(state.estimates) < 2:
        content.append("You need at least two estimates to create a comparison.")

    else:
        names = []
        costs = []
```

```
        for project in state.estimates:
            names.append(project.name)
            costs.append(project.estimated_cost)

        plt.figure(figsize=(8, 5))
        plt.bar(names, costs, color="steelblue")

        plt.title("Paving Project Cost Comparison")
        plt.xlabel("Projects")
        plt.ylabel("Estimated Cost ($)")
        plt.xticks(rotation=30)
        plt.tight_layout()

        content.append(Text("Compare the estimated costs of your projects:"))
        content.append(MatPlotLibPlot())

    content.append(Button("Back to Estimates", "estimates_page"))

    return Page(state, content)
```

3. What the graph will look like

For example, if the user has entered three estimates:

Paving Project Cost Comparison

Example only — estimated cost in dollars

$0$2K$4K$6K$8K

Driveway

Parking Area

Sidewalk

Driveway: Estimated cost $3K
Parking Area: Estimated cost $6.5K
Sidewalk: Estimated cost $1.8K

4. How this helps meet your project requirements

- Loop: Goes through `state.estimates` to collect project names and costs.
- If statement: Checks whether at least two estimates exist.
- Matplotlib: Creates the bar graph.
- State: Uses the estimates created during the current session.

One note: Matplotlib may take a little longer to load the first time because Drafter installs it when needed. Also, the graph is an image, not an interactive chart.